FLINK FORWARD #ASIA 2021 ONLINE

实时即未来 REAL-TIME IS THE FUTURE

# 基于 Flink Dynamic Table 构建流批一体数仓

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#1 业务需求

#2 业务痛点

#3 动态表

#4 流式数仓

#5 **PLAN**

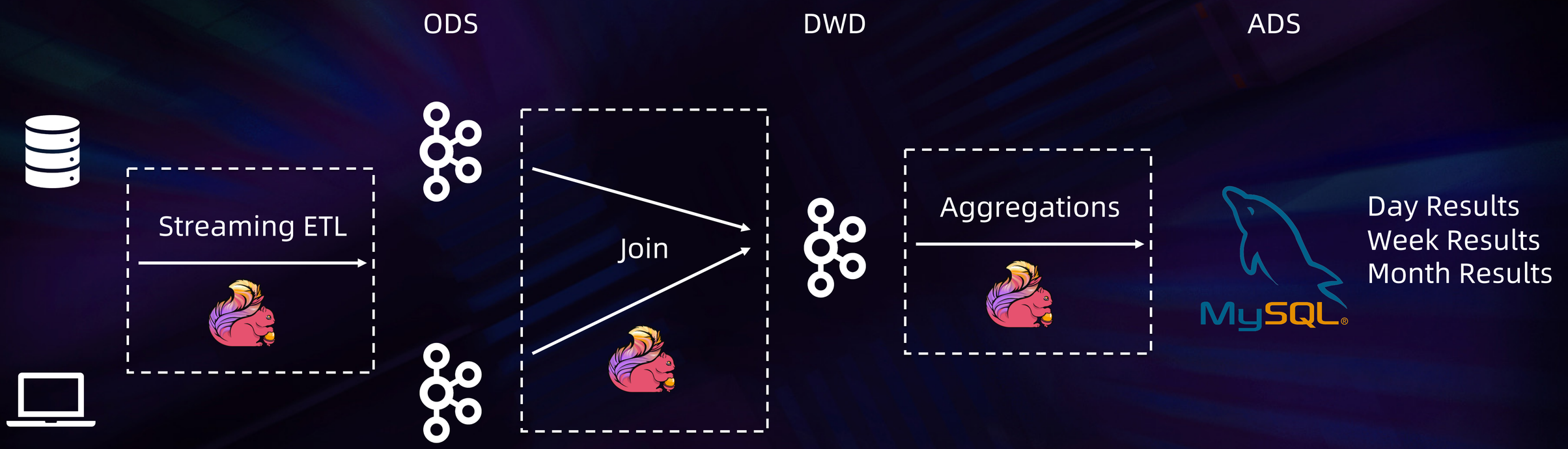
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# #1 业务需求

# 搭建实时数仓

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- BOSS: 快快快！给我看到实时的数据！



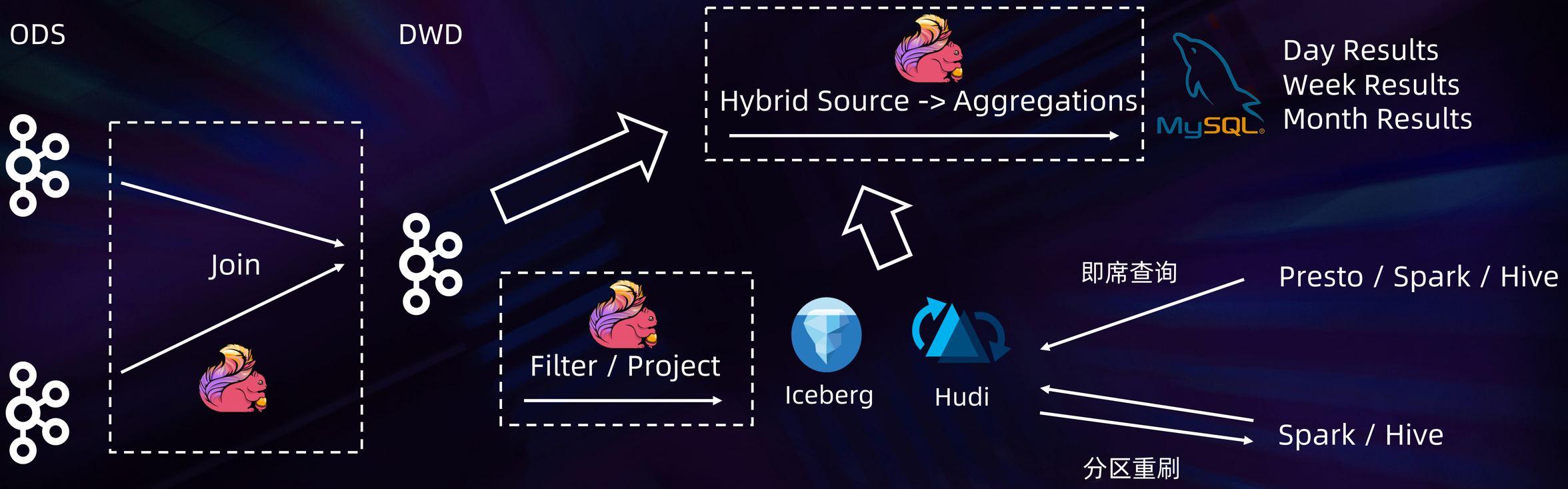
# 即席查询与数据订正

- BOSS: 我能不能随便查查数仓的数据?
- BOSS: 有脏数据? 清理下?



# 新增实时业务指标

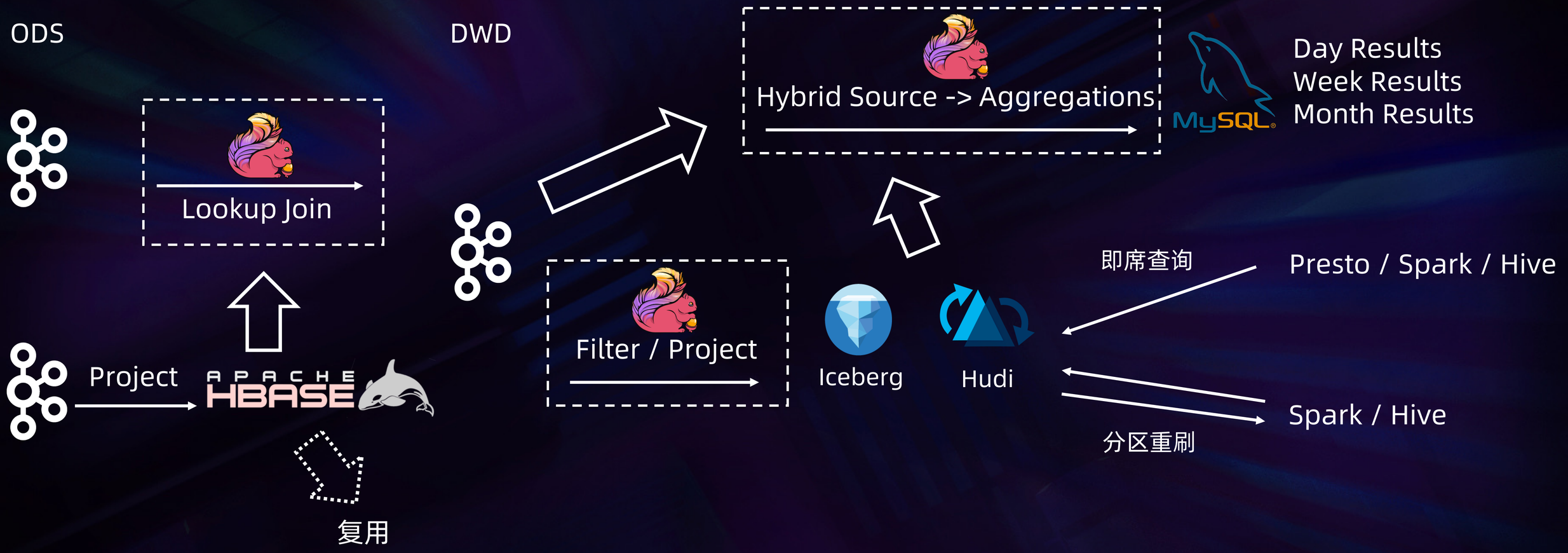
- BOSS: 来，给我新增一些业务指标！

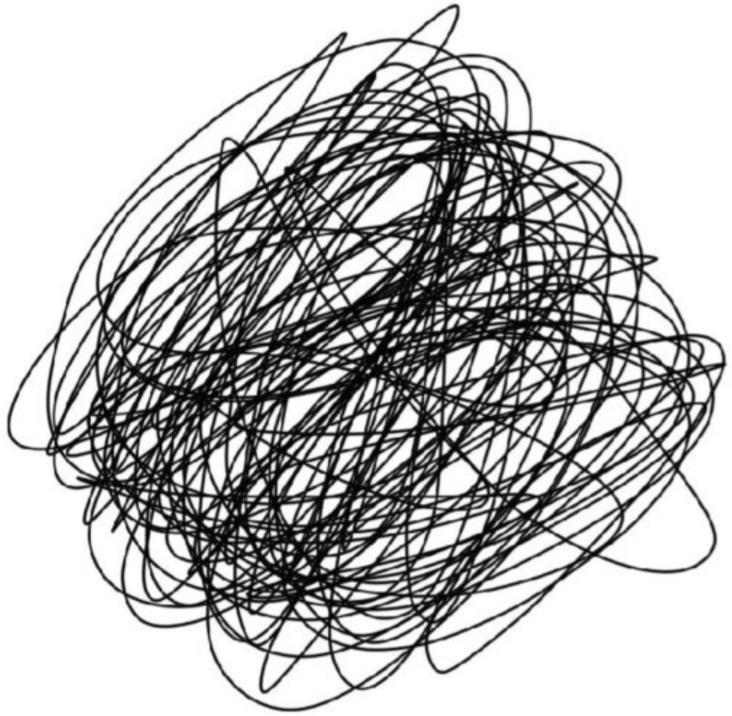


# 维表节省成本

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- BOSS: 成本太高了，给我省着点！






我们真的需要这么多系统吗？

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# #2 业务痛点

# 实时数仓痛点

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- Kafka 成本昂贵
  - Kafka 只保留最近数据
  - 导出到 Hive？系统复杂度
- Kafka 难以查询
  - 行存：几乎不可查询
  - 导出到 KV engine / OLAP，系统复杂度
- 流计算 难以使用
  - 各种 Changelog Kind，增量计算模型？
  - “sink only accepts append-only messages”
  - 数据怎么多了？手动去重？
- 流批一体 难以达成
  - 流批割裂，数据、表结构和计算越行越远
  - 流批难以协作，比如 backfilling、流连接批维表



# workaround：中间数据导出去




带给用户的复杂：

- Append Only & Upsert
- Primary Key & Timestamp
- 依赖 State
- 流与批 不一致的体验

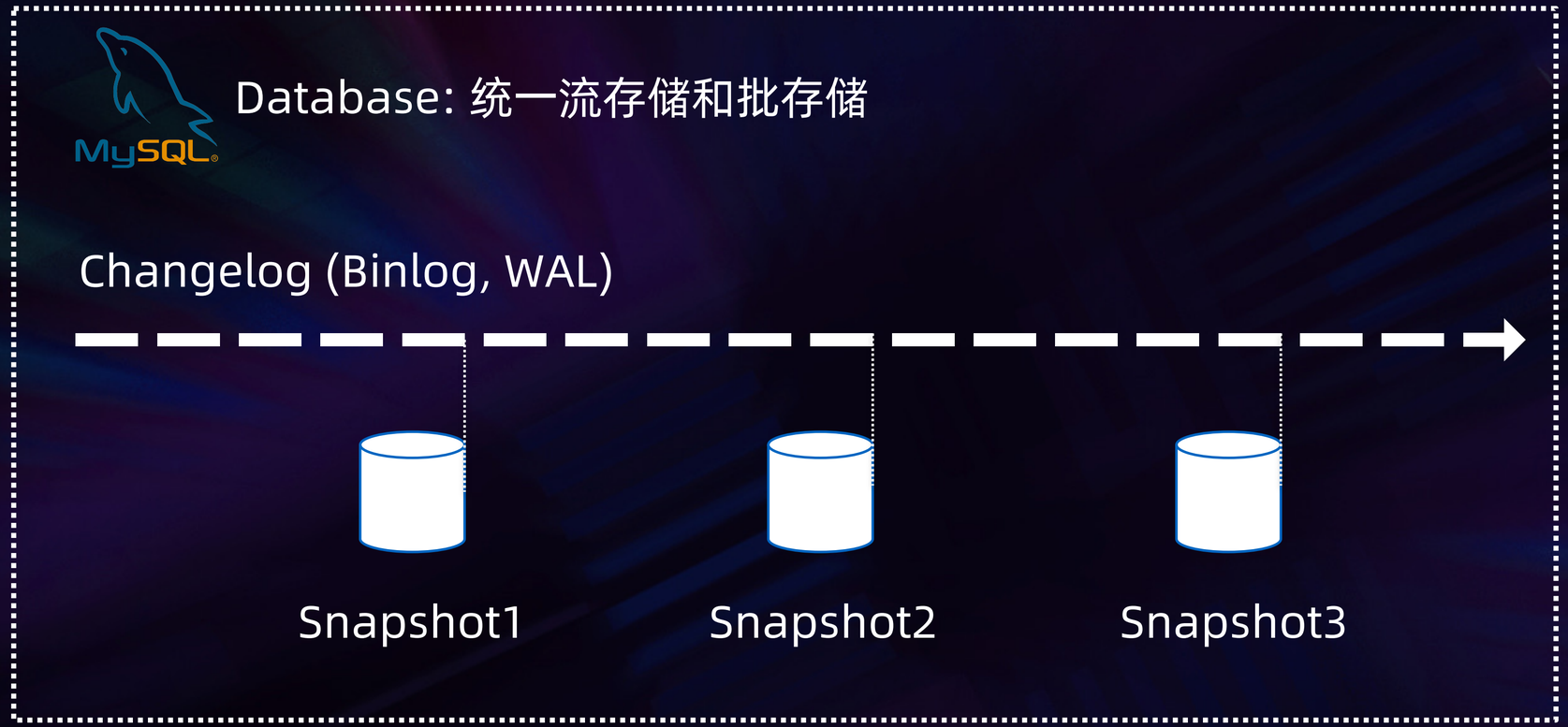
带给平台的复杂：

- 运维成本：Maintain Servers
- 学习成本：用法各不相同
- 打通成本：不同系统难以打通

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# #3 动态表

# Rethink 流和批




数据仓库的流批一体存储：Database 的体验！

# 动态表：全新的Flink内置存储

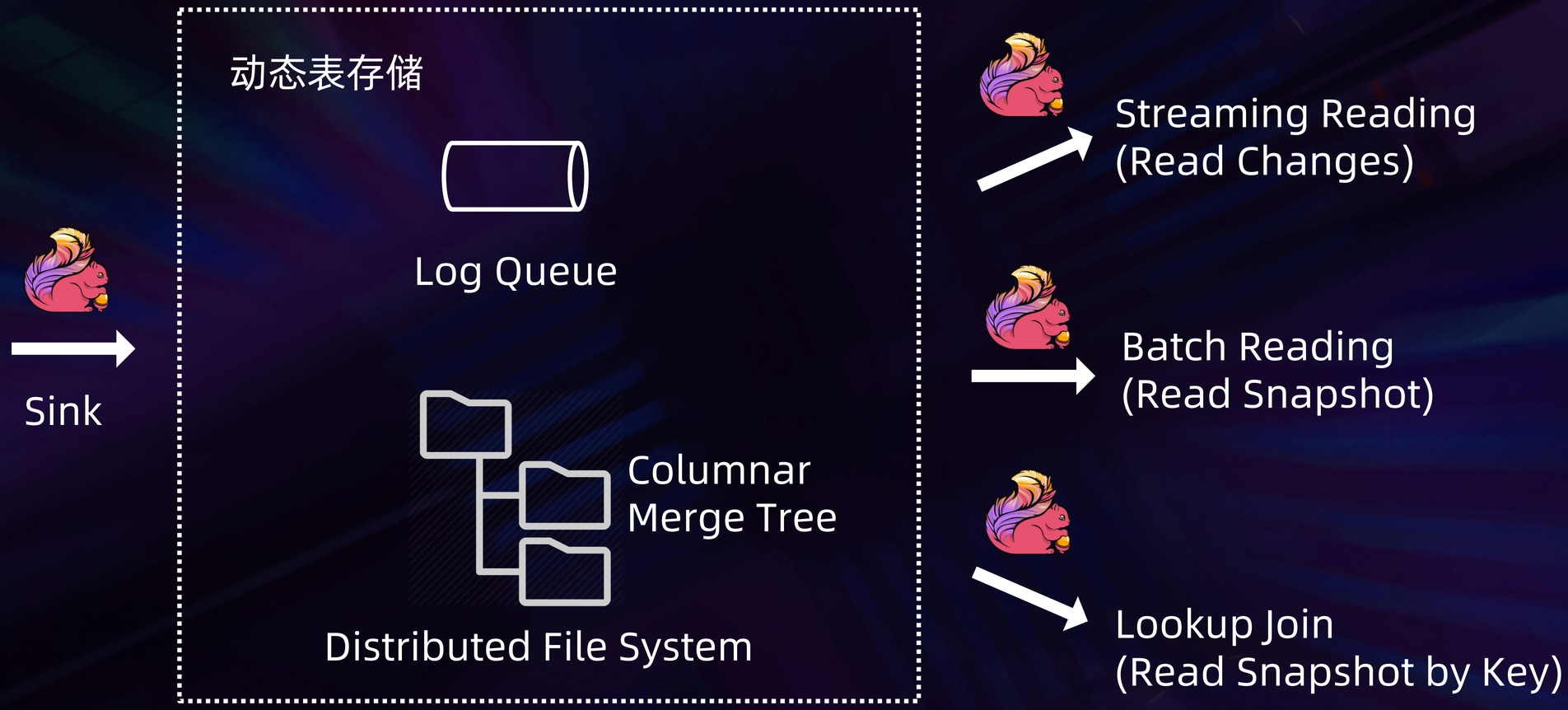
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Flink 动态表

- Snapshot + Log
- 满足所有<实时> User Case
- 存储易用 直接查询DFS

# 动态表：存储结构


动态表存储
Log Queue
Columnar
Merge Tree
Distributed File System
Sink
Streaming Reading
(Read Changes)
Batch Reading
(Read Snapshot)
Lookup Join
(Read Snapshot by Key)

# 动态表：流批一体

- 读取
  - 流读：读取 Log Changes
  - 批读：读取 Snapshot
  - 流批融合：Hybrid 读
  - Lookup：支持点查
- 写入
  - 流写：持续插入
  - 批写：支持分区、支持Overwrite
- Lake Store
  - On DFS，无服务，计算存储分离
  - 列存高性能分析



# 动态表：全面支持 SQL

- 支持任意 DDL 定义
  - 只有业务逻辑
  - 主键定义可选
  - 支持所有类型
- 支持任意 SQL 插入
  - 任意消息类型
  - 全面DML支持

```
CREATE TABLE user_behavior (
  user_id BIGINT,
  item_id BIGINT,
  category_id BIGINT,
  behavior STRING,
  dt STRING,
  ...,
  PRIMARY KEY (user_id, item_id) NOT ENFORCED
) PARTITIONED BY (dt);
```

Optional　　不用定义 Options　　Optional

```
INSERT INTO user_behavior SELECT ...
```

支持任意 SQL 的写入

# 动态表：Merge Tree

- LSM: Log-Structured Merge Tree
- 支持大量快速更新
  - 更新写增量文件
  - 基于 Sorted File 的按需 Merge
- 支持高效分析
  - 全局有序 Sorted File 的 Data Skipping
  - Sort Aggregation and Sort Merge Join
- 支持点查

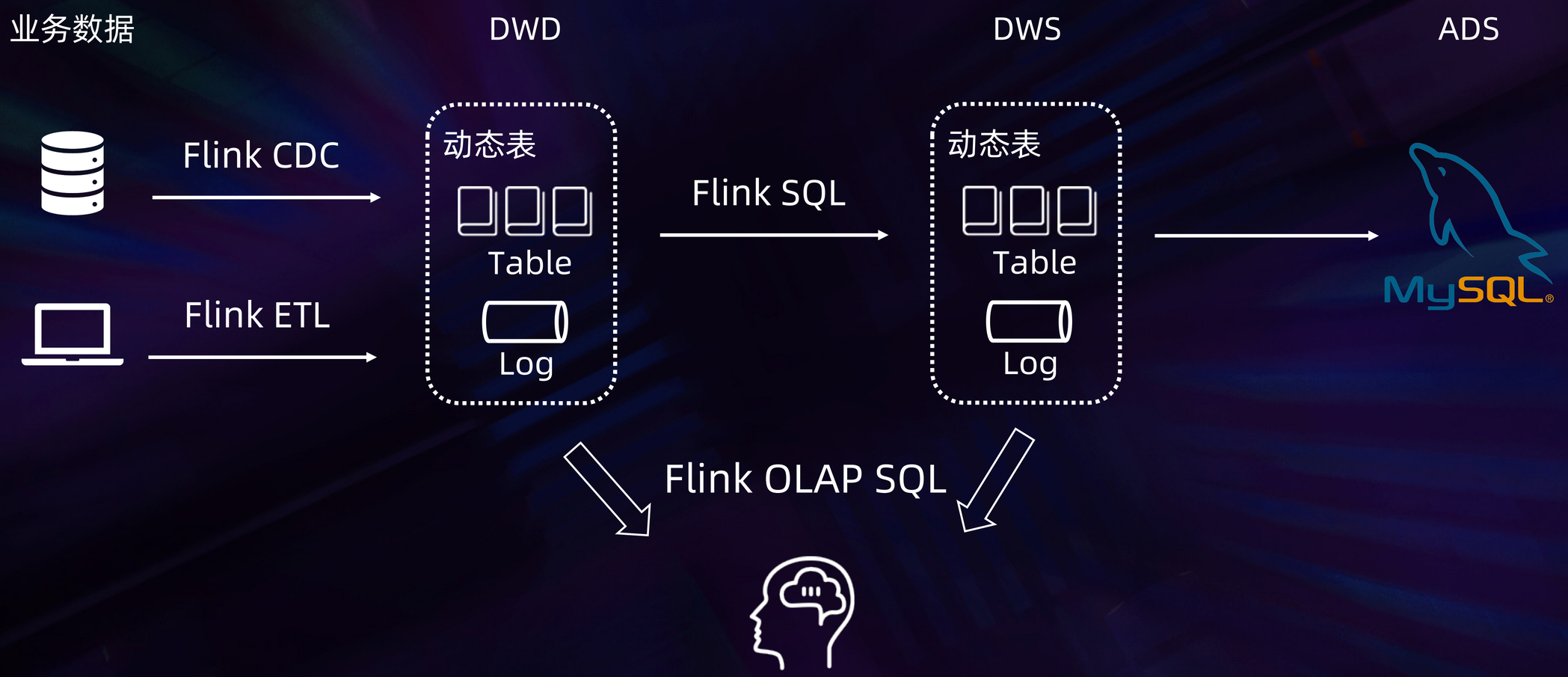
**Merge Tree 是实时的核心**



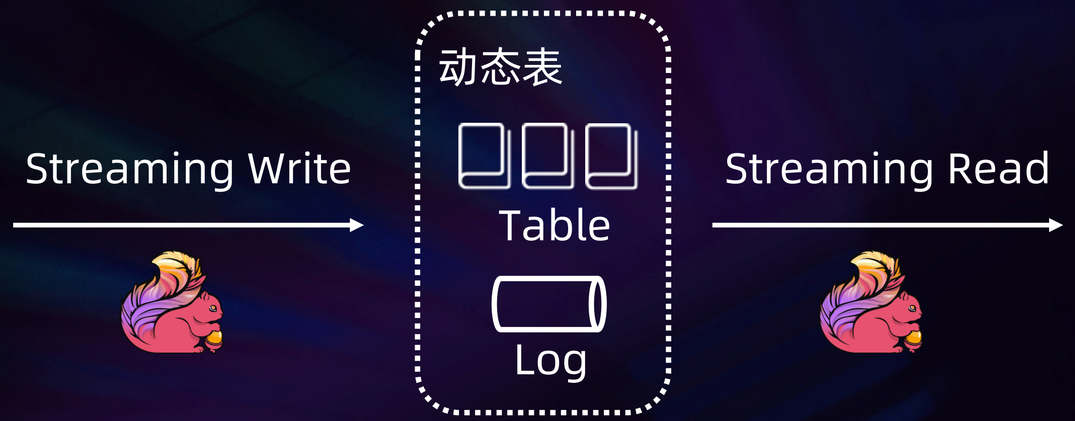
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# #4 流式数仓

# 流式数仓：Streaming Warehouse



# 流式数仓：流写流读




最好用的 Log Connector

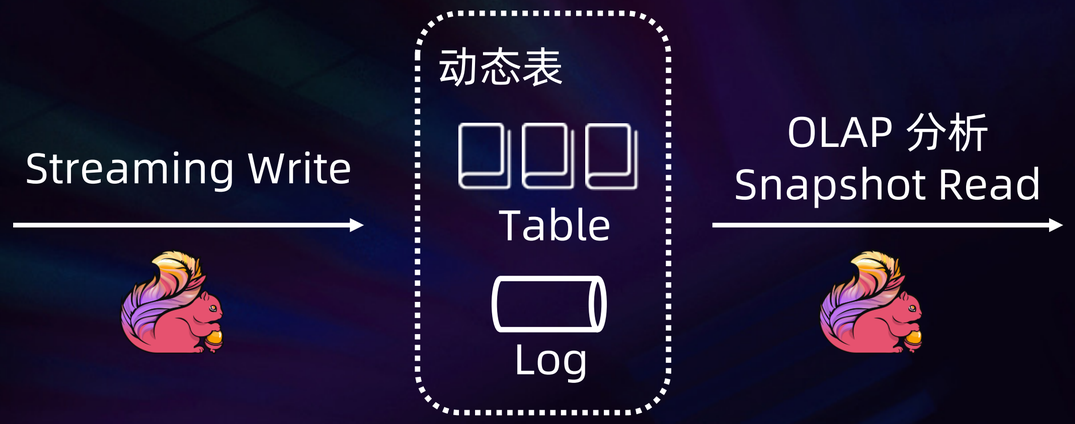
- 实时：最低毫秒级延迟
- 准确：Exactly-once
- 全面：输入支持Changes
- 优化：减少物化

```
SET ‘execution.runtime-mode’ = ‘streaming’;

INSERT INTO user_behavior SELECT ...

SELECT * FROM user_behavior;
```

# 流式数仓：OLAP

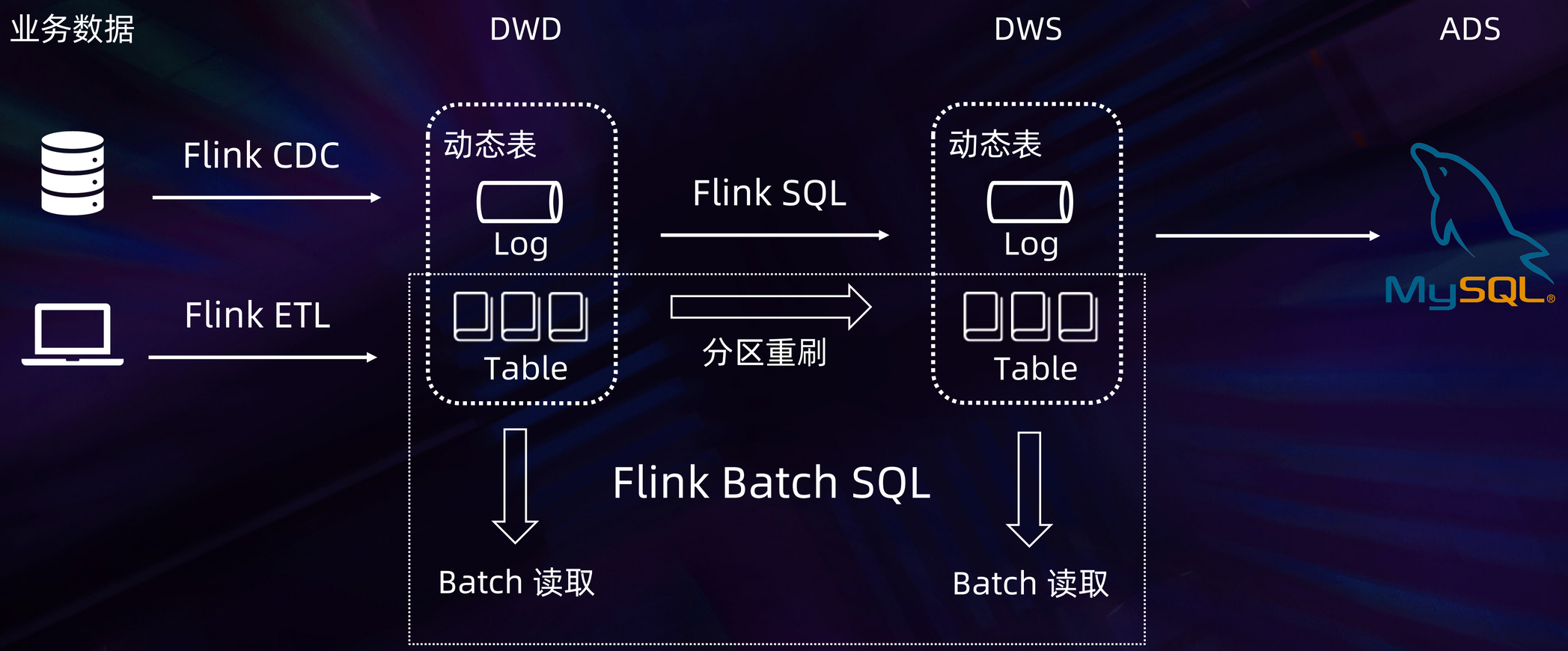



- OLAP 查询实时数据
- OLAP 查询历史数据
- 快速：强大的 Data Skipping
- 准确：保存所有数据
- 灵活：支持分布式 Join、聚合、Over等

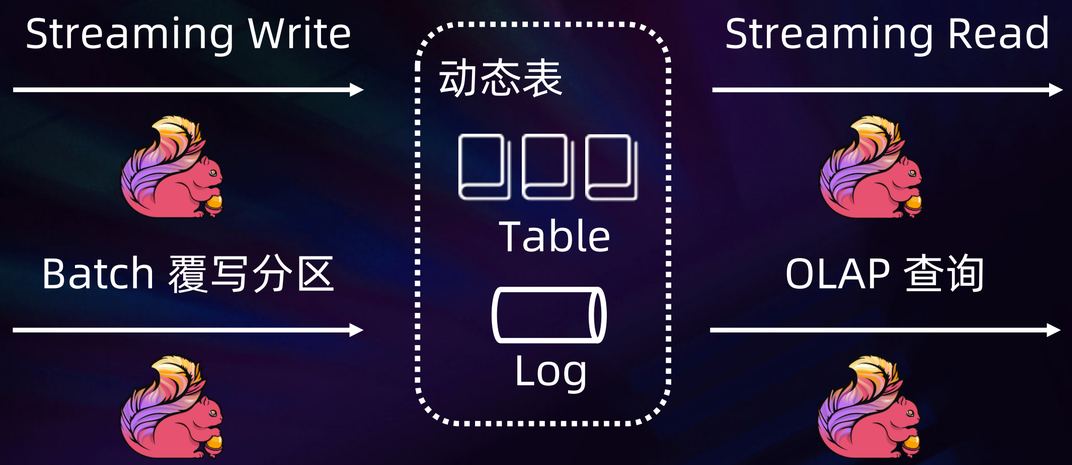
```
SET ‘execution.runtime-mode’ = ‘batch’;

SELECT address, COUNT(*) AS cnt
FROM user_behavior b JOIN user_attribute a
ON b.user_id = a.user_id
GROUP BY b.address;
```

# 流式数仓-离线：分区重刷


业务数据
DWD
DWS
ADS
Flink CDC
Flink ETL
动态表
Log
Table
Flink SQL
分区重刷
动态表
Log
Table
MySQL
Flink Batch SQL
Batch 读取
Batch 读取

# 流式数仓-离线：数据订正

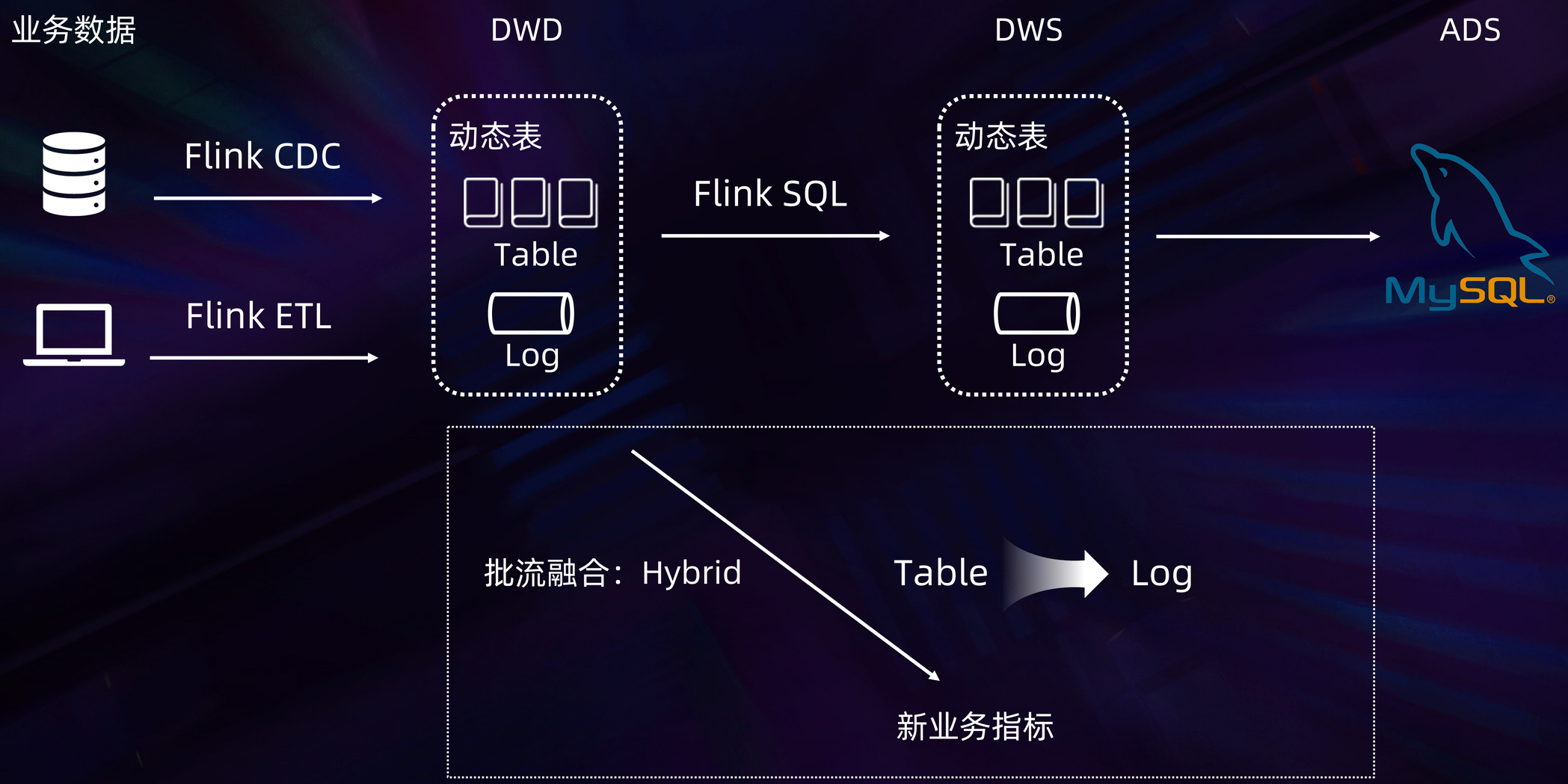



- ACID：并发写入，流批互不干扰
- 全面：支持各种 Batch DML
- 当做离线数仓构建 Batch Pipeline

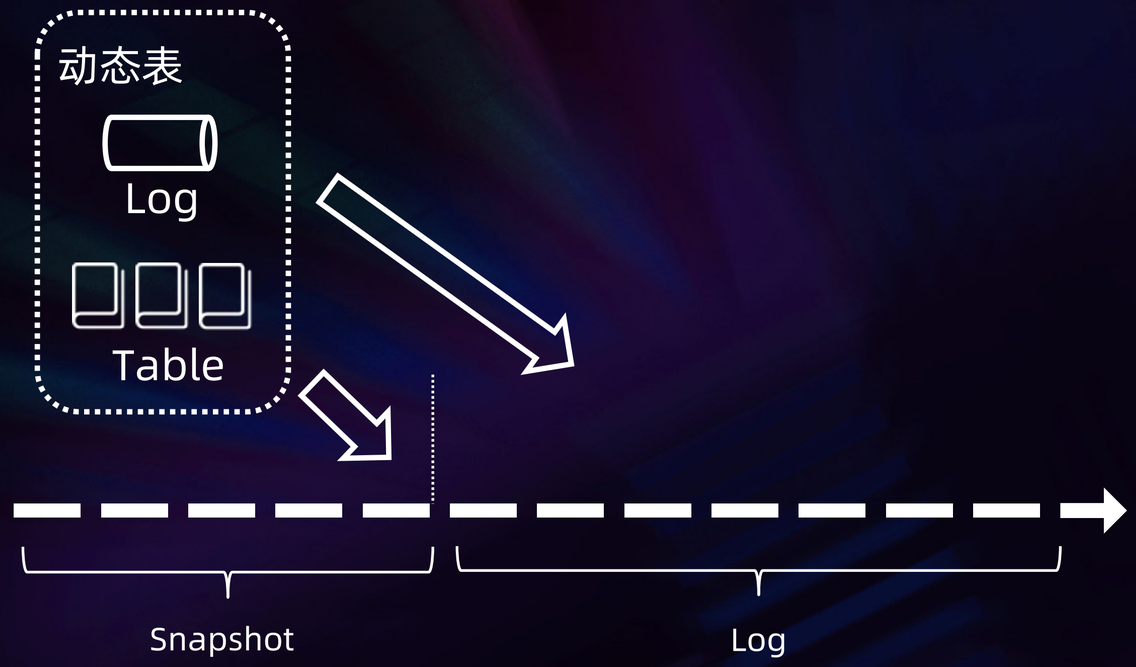
```
SET ‘execution.runtime-mode’ = ‘batch’;

INSERT OVERWRITE user_behavior PARTITION (dt=‘20211130’)
SELECT ...;
```

# 流式数仓：上线新业务


业务数据
DWD
DWS
ADS
Flink CDC
Flink ETL
动态表
Table
Log
Flink SQL
动态表
Table
Log
MySQL
批流融合：Hybrid
Table
Log
新业务指标

# 流式数仓：新增实时指标




- Exactly-once，严格对齐
- 全量 + Partition Pruning => 增量

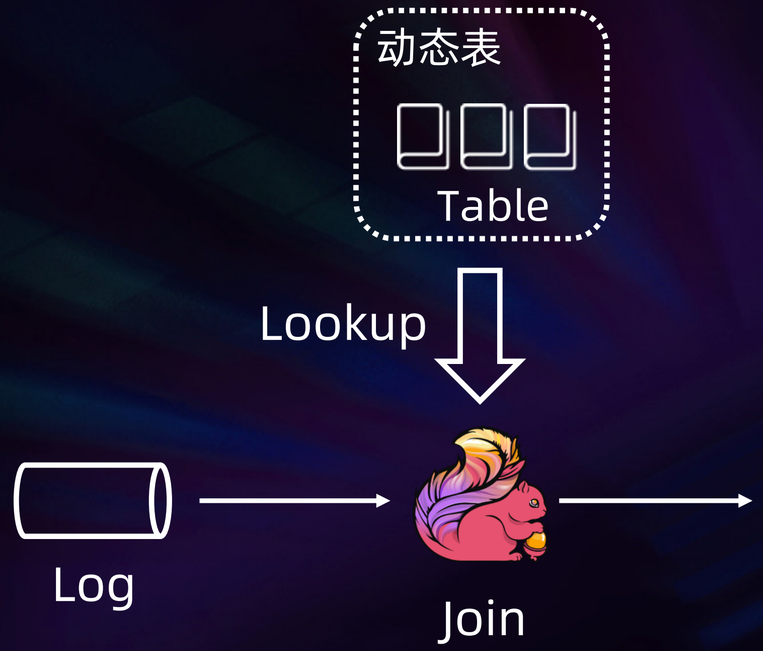
```
SET 'execution.runtime-mode' = 'streaming';

SELECT * FROM user_behavior /*+OPTIONS('log.scan'='full') */
WHERE dt >= 20211130;
```

log scan 支持：
1.full (默认)
2.latest
3.from-timestamp

Partition Prune

# 流式数仓：高性能 Join




- Table 无服务
- 存储计算分离：点查 LSM
- 三级存储：内存、本地磁盘和 DFS
- Watermark 对齐

```
SET ‘execution.runtime-mode’ = ‘streaming’;

SELECT * FROM my_stream LEFT JOIN user_behavior FOR SYSTEM AS OF
my_stream.proctime ON id = user_id;
```

# Demo演示

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# #5 PLAN

# PLAN

- Flink 1.15  推出 MVP 版本
- 后续支持 Lookup
- 后续支持 Join Engine

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提供专项支持


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